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Wearable near-infrared spectroscopy device for low-resource settings

Saeed Samaei^{1,2,*} , Rasa Eskandari¹ , Leena N Shoemaker^{1,2,3} , Natalie C Li⁴ , Farah Kamar^{1,2} , Lee D Sikstrom^{1,5} , David W Holdsworth^{1,4,5,6} and Mamadou Diop^{1,2,4,6}

¹ Department of Medical Biophysics, Schulich School of Medicine and Dentistry, Western University, London, Ontario, Canada

² Imaging Program, Lawson Research Institute, London, Ontario, Canada

³ School of Kinesiology, Western University, London, Ontario, Canada

⁴ School of Biomedical Engineering, Western University, London, Ontario, Canada

⁵ Roberts Research Institute, Western University, London, Ontario, Canada

⁶ Frugal Biomedical Innovations Program, Western University, London, Ontario, Canada

* Author to whom any correspondence should be addressed.

E-mail: ssamaei@uwo.ca and mdiop@uwo.ca

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Supplementary material for this article is available [online](#)

Abstract

There is a global need for accessible neuromonitoring technologies in low-resource settings. A low-cost, wearable near-infrared spectroscopy device was developed using a commercially available optical biosensing module. The device features wireless data transmission, dual-wavelength illumination, and supports dual-slope analysis for improved brain sensitivity. Its performance and depth sensitivity were validated through layered blood phantoms and *in vivo* experiments, demonstrating stable operation, reliable signal quality, and accurate detection of cerebral oxygenation changes in adult head mimicking phantoms and healthy adult volunteers. The device is low-cost, reproducible with minimal engineering skills, easy to use, and reliable for *in vivo* human experiments, highlighting its potential as a practical tool for non-invasive brain monitoring, particularly in low-resource environments. The data-processing codes and the 3D-printable probe design files have all been open-sourced and are publicly available.

1. Introduction

Low and middle-income countries (LMICs) bear a disproportionate burden of neonatal brain injury, primarily due to birth asphyxia, infections, and inadequate perinatal care [1–3]. While neuromonitoring can detect impairment, guide interventions, and predict outcomes, healthcare systems in LMICs face a critical shortage of accessible neuromonitoring tools [3]. Conventional brain monitoring technologies, such as Transcranial Doppler ultrasound, Positron Emission Tomography, and Magnetic Resonance Imaging, are prohibitively expensive, require highly trained personnel, and depend on infrastructures that are typically unavailable in resource-limited settings. As a result, their widespread adoption in LMICs remains limited. Despite the urgent need, frugal and accessible solutions for neurological evaluation are surprisingly scarce. This highlights the importance of developing neuromonitoring tools that are affordable, easy to use, and operable by healthcare workers with minimal technical training. Moreover, reducing cost and complexity is essential in under-resourced environments, where advanced equipment and specialized personnel are often lacking. To be practical in these settings, such systems must be battery-powered, as access to reliable electricity is often inconsistent [4].

Near-infrared spectroscopy (NIRS) is a non-invasive technique that can provide continuous, bedside monitoring of cerebral hemodynamics. It leverages the distinct absorption spectra of tissue chromophores, such as oxygenated hemoglobin (HbO₂) and deoxygenated hemoglobin (Hb), to quantify

Table 1. Open-source^a wearable NIRS device comparison.

NIRS device	Maximum channel number ^b	Wavelengths (nm)	Sampling rate	SDD (mm)	Dual-slope measurement ^c	Cost ^d
HEGduino [15, 21]	2	660, 880 & 940	64 Hz	30	No	\$ ^e
FlexNIRS [22]	6	735 & 850	100 Hz	8/28/33	Yes	\$\$
DIY-fNIRS [23]	4	740 & 850	10 Hz	5/10/23/28	No	\$\$
OpenNIRScap [24]	24	660 & 940	1 kHz	Short and long ^f	No	\$\$
NIRDunio [25]	128	740 & 860	5 Hz	8–30	No	\$\$\$
OpenNIRS [26]	4	750 & 850	15 Hz	35	No	N/A

^a All listed modules provide open-source resources, accessible through the associated publications.

^b Each channel includes one detector and at least two emitters operating at different wavelengths.

^c The dual-slope approach improves depth sensitivity and is described in the Data Processing section.

^d Cost categories are represented as: '\$' (<US\$100), '\$' (US\$100–500), and '\$' (> US \$500). Values reflect component-level costs reported in the associated publications.

^e Value may reflect the price of a fully mounted device.

^f The SDDs were not specified in the original publication. SDD denotes source–detector distances.

changes in their concentrations [5–7]. These measurements can be used to compute tissue oxygen saturation (StO₂), which reflects the balance between oxygen supply and demand in the microvasculature [5, 8]. Over the past decade, NIRS-based technologies have been increasingly employed to monitor StO₂ in intensive care and surgical procedures [6, 9–11]. The technique is also widely adopted for neonatal neuromonitoring [12–14], and has potential for early detection of brain injuries in low-resource settings [3, 6].

Currently, most NIRS devices developed by research labs are fiber-based, bulky, and rely on custom hardware, making them impractical for low-resource environments [15]. Interestingly, recent advances in optoelectronics and signal processing have enabled simpler commercial cerebral oximeters that require minimal user training and provide continuous, real-time StO₂ readings [16–18]. These innovations highlight the potential of NIRS as an accessible monitoring tool. Nevertheless, current commercial systems remain costly and often require continuous external power, further limiting their practicality in low-resource settings [4]. Recently, low-cost NIRS systems have been introduced to address some of these limitations [19, 20]. Table 1 summarizes the main characteristics of available open-source wearable NIRS systems.

The majority of these open-source wearable NIRS systems use conventional single-source-detector acquisition schemes, which provide limited depth sensitivity and are highly affected by extracerebral-layer contamination [27]. Wu *et al* [22] recently introduced FlexNIRS, a wearable, battery-operated NIRS system designed to reduce costs, and notably incorporating the dual-slope (also called 'self-calibrated') measurement approach that improves depth sensitivity compared to conventional designs [27]. While promising, these devices rely on custom hardware, requiring specialized assembly and maintenance, which limits their reproducibility and scalability by end users in LMICs. This highlights the ongoing need for more accessible and easily deployable alternatives. Interestingly, consumer-grade biosensors and smartwatches are now widely used for everyday monitoring of heart rate and arterial oximetry [28]. These devices already contain the necessary optoelectronics components for cerebral oximetry and could be repurposed for low-cost, non-invasive brain monitoring with minimal engineering effort.

This work aimed to evaluate the feasibility of adapting a commercially available open-source optical biosensing module for cerebral tissue oximetry. Our approach repurposes key electronics and optical components of the module to construct a wearable, wireless NIRS system. The resulting device is low-cost, can be reproduced with basic electronic assembly skills, and is capable of continuous monitoring of hemoglobin concentrations, StO₂, and heart rate. The emitter-detector arrangement enables the system to take advantage of the enhanced depth sensitivity provided by dual-slope measurements. We characterized the device's performance and its depth sensitivity through two-layer tissue-mimicking blood phantoms and a series of *in vivo* experiments in which healthy adults underwent carotid compression (CC) and hypercapnia challenges. The preassembled evaluation module used in this study is commercially available, resulting in a NIRS device that can be fabricated with minimal technical complexities. The required modifications are limited to replacing the factory-installed LEDs with appropriate NIR

emitters and mounting the module onto a 3D-printed probe, allowing it to be reproduced and operated by non-experts. This approach offers a promising tool for point-of-care neonatal brain monitoring in low-resource settings. To support reproducibility, the data-processing scripts and 3D-printable probe design files have been open-sourced and are publicly available.

2. Method

2.1. Hardware development

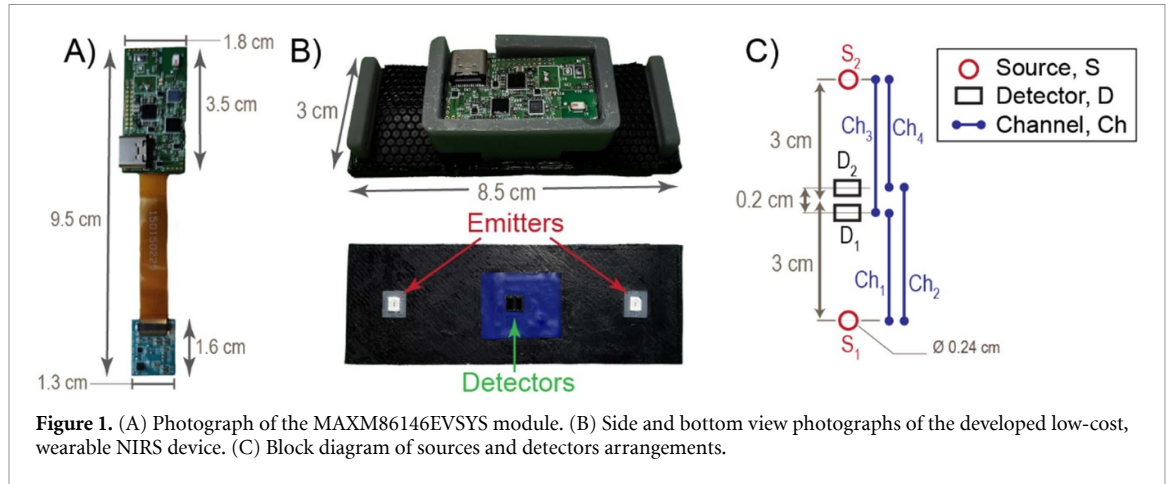
We developed a dual-wavelength, four-channel continuous-wave (CW) NIRS device based on the electronics of a commercially available optical biosensing module (MAXM86146EVSYS, Maxim Integrated). This consumer-grade photoplethysmography (PPG) module offers several built-in capabilities, such as a robust noise-canceling unit, several LED drivers, and a pair of sensitive detectors, making it well-suited for our aim. By taking advantage of these features, the requirement of designing and building custom circuits was eliminated, allowing us to focus on enhancing depth sensitivity for brain monitoring. Notably, a major challenge with NIRS is signal contamination from superficial tissues [29]. However, recent advances in data processing [30–32], such as the dual-slope method [27, 33], improve cerebral sensitivity by comparing light attenuation at different source-detector distances (SDDs). The MAXM86146EVSYS module's flexible configuration, supporting multiple LED drivers and detector channels, allowed us to implement this approach to improve sensitivity to deeper tissue. The resulting wearable NIRS device is depicted in figure 1, and its characteristics are provided in the following sections.

Emitters: the MAXM86146EVSYS module is equipped with four programmable LED drivers, allowing independent control of each light source. Originally, the module was equipped with two green LEDs (536 nm) and an LED package containing two emitters (660 nm and 950 nm). To adapt the device for cerebral oximetry, these factory-configured emitters were replaced with a pair of dual-wavelength LEDs (SMT735D/850D, Marubeni) emitting at 735 nm and 850 nm. A flexible 3D-printed support was used to position each LED package 3 cm from the center of the probe, where the detectors are located. The system assembly procedure is described in the supplementary material. As shown in figure 1(C), the emitters and detectors were arranged symmetrically, which enabled implementation of the dual-slope approach [27, 30]. More details on the data analysis related to this approach are provided in the Data Processing section. The emitter pulse width and output power are adjustable via the manufacturer-provided software interface. The light pulse width can be set to 14.8, 29.4, 58.7, or 117.3 μs . Also, each LED can be driven with a tunable forward current ranging from 0 mA (off) to 120 mA. This corresponds to up to 50 mW peak power per wavelength distributed over a 2.4 mm diameter illumination spot on the sample surface.

Detectors: the PPG module includes two integrated PIN photodiodes, separated by 2 mm. These photodiodes have a peak sensitivity at 860 nm, with a spectral bandwidth range from 420 to 1020 nm. Further, the photodiodes maintain linearity across a wide dynamic range, providing reliable detection of intensity variations caused by physiological changes in tissue. The photodiodes also have a large active area (each 3.8 mm²), which enhances photon collection efficiency.

Data acquisition and ambient light cancellation: the MAXM86146EVSYS enables low-noise optical signal acquisition at a fixed sampling frequency of 25 Hz. To improve signal quality, the module is equipped with built-in analog front-end (AFE) electronics that reduce ambient light contamination through a time-multiplexed sequencing strategy. While the detectors continuously receive light, the four LEDs are sequentially activated for short durations based on user-defined pulse width and sampling rate settings. The intensity of diffusely reflected light at each time point can be acquired either as a single-shot measurement or averaged over multiple samples. The module supports various acquisition modes, ranging from 25 samples per second (sps) with no averaging to 400 sps with 16-sample averaging.

At the optimal setting, as described in section 3.1, the system acquires data at 400 sps with 16-sample averaging, using a 58.7 μs pulse width and a 120 mA driving current. In this configuration, it operates at a 25 Hz acquisition rate (25 cycles per second), with each LED turned on 16 times per cycle. During each cycle, an LED is active for 939.2 μs , corresponding to a duty cycle of 2.4% per LED. This results in a time-averaged illumination power of 1.2 mW for each LED, distributed over an illumination area with a 2.4 mm diameter on the sample. This corresponds to an irradiance of 0.27 mW mm⁻², which is well below the American National Standards Institute safety limits for skin exposure in the NIR range (2.7 – 4 mW mm⁻²) [34].



The module's embedded noise-cancellation algorithm treats signals collected without LED illumination as ambient light and subtracts them from data recorded with the LED on. This approach significantly reduces ambient light contamination and improves signal quality prior to digitization. Subtracting ambient noise before digitization also prevents it from reaching the analog-to-digital converter (ADC) threshold and helps preserve dynamic range. The corrected signals are digitized using a high-resolution 19-bit ADC, temporarily stored in a buffer, and then transferred to an external device via Bluetooth.

Additional features of the system: the developed NIRS device operates on a rechargeable battery, supporting up to 4 h of continuous data acquisition on a full charge. It features integrated Bluetooth low energy for wireless data transmission, optimized for low power consumption, which is ideal for wearable applications. The device is also equipped with a built-in 3-axis accelerometer that enables synchronized motion monitoring alongside optical measurements.

The module's software enables users to adjust key acquisition parameters, including LED sequence, emitters' drive current, illumination pulse width, sampling rate, and averaging modes. It displays ambient-noise-corrected light intensity in real time across eight optical channels (four LEDs and two detectors). The data acquisition procedure is described in the supplementary material. The complete system is compact and low-cost, with a total hardware cost below \$200 (USD).

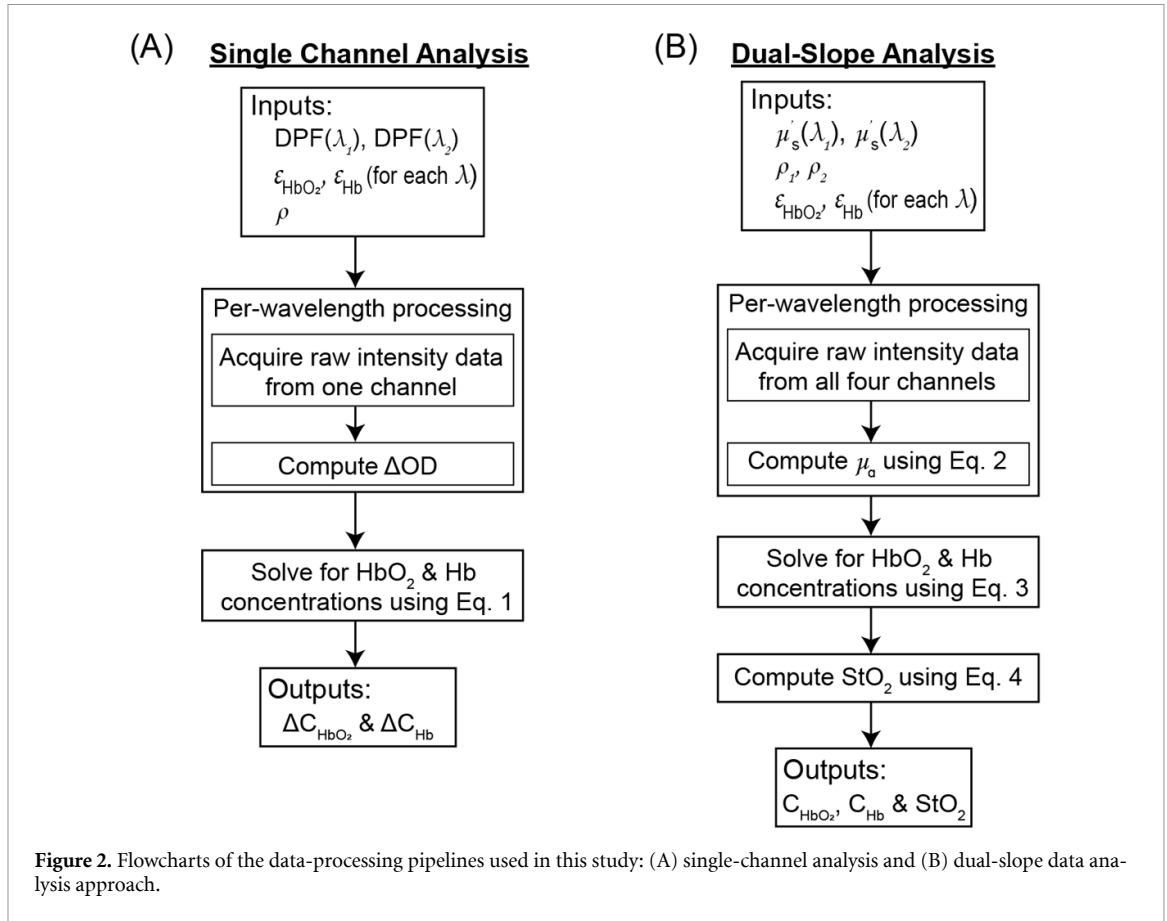
2.2. Data processing

The arrangement of the LEDs and detectors (figure 1(C)) enables two data-processing approaches, single-channel and dual-slope analysis, to be applied to the recorded raw signals, as illustrated in figure 2.

Single-channel analysis: a conventional approach in NIRS is to estimate changes in concentration of oxy-hemoglobin, C_{HbO_2} , and deoxy-hemoglobin, C_{Hb} , using the modified Beer-Lambert Law (MBLL) [35, 36], based on measurements from a single source-detector pair. In this configuration, the two wavelengths are emitted sequentially, and the diffusely reflected light is collected by a detector positioned a few centimeters away from the source. The MBLL calculation, presented in equation (1), requires the molar extinction coefficients, ϵ , of HbO_2 and Hb at both wavelengths in order to derive changes in their concentrations from variations in the detected light intensity. Wavelength-specific values of the molar extinction coefficients were sourced from [37],

$$\begin{bmatrix} \Delta C_{HbO_2} \\ \Delta C_{Hb} \end{bmatrix} = \left(\rho \cdot \left(\begin{bmatrix} \epsilon_{HbO_2, \lambda_1} & \epsilon_{HbO_2, \lambda_2} \\ \epsilon_{Hb, \lambda_1} & \epsilon_{Hb, \lambda_2} \end{bmatrix} \times \begin{bmatrix} DPF_{\lambda_1} & 0 \\ 0 & DPF_{\lambda_2} \end{bmatrix} \right)^T \right)^{-1} \times \begin{bmatrix} \Delta OD_{\lambda_1} \\ \Delta OD_{\lambda_2} \end{bmatrix} \quad (1)$$

where ΔC indicates changes of concentration; ρ is SDD; DPF_{λ_i} is differential pathlength factors for wavelength λ_i . Wavelength-specific values of DPF are reported in [38]. Specifically, the DPF values used in this study were 6.2 at 735 nm and 5.8 at 850 nm. ΔOD represents changes in optical density, defined as $-\log(I_{final}/I_{initial})$, where $I_{initial}$ and I_{final} correspond to the intensities recorded before and after the concentration change [36]. The superscript T denotes the transpose of the matrix. While this approach allows tracking temporal variations in tissue hemodynamics, it has limited depth-sensitivity.



Dual-slope analysis: multi-distance measurements in a symmetric geometry enable the use of the dual-slope processing strategy to enhance depth sensitivity [27, 30]. This approach estimates the effective attenuation coefficient (μ_{eff}), which depends on both the absorption and scattering properties of the tissue, based on light intensities measured at different SDDs. The slope of the logarithmic intensity versus SDD is proportional to μ_{eff} ($\sim \sqrt{3\mu_a\mu'_s}$), where μ_a represents absorption coefficient and μ'_s indicates the reduced scattering coefficient of the sample [39]. By assigning a value to μ'_s , μ_a can be estimated for each wavelength, as defined in equation (2) [22],

$$\mu_a(\lambda_i) = \frac{1}{3\mu'_s(\lambda_i)} \left(\frac{1}{\rho_1 - \rho_2} \ln \left(\left(\frac{\rho_2}{\rho_1} \right)^2 \times \sqrt{\frac{I(\rho_2, S_{1\lambda_i}, D_2) \times I(\rho_2, S_{2\lambda_i}, D_1)}{I(\rho_1, S_{1\lambda_i}, D_1) \times I(\rho_1, S_{2\lambda_i}, D_2)}} \right) \right)^2 \quad (2)$$

where $I(\rho_i, S_j, D_k)$ corresponds to the light intensity detected by detector D_k at a separation ρ_i from emitter S_j . For example, the light intensity measured using channel 3, labeled as Ch. 3 in figure 1(C), is denoted as $I(\rho_2, S_2, D_1)$ in equation (2).

The μ'_s values can be measured using other NIRS modalities, such as time-resolved (trNIRS) [40] or frequency-domain (FD-NIRS) [30], or adopted from the literature [30, 40–42]. To enhance the accuracy of our *in vivo* analysis, we experimentally determined μ'_s values using trNIRS measurements, as described in section 2.6. The estimated μ_a values were then used to compute the absolute concentrations of HbO_2 and Hb by solving the system of equations for the molar concentration of these chromophores (equation (3)) [36],

$$\mu_a(\lambda_i) = \epsilon_{HbO_2, \lambda_i} C_{HbO_2} + \epsilon_{Hb, \lambda_i} C_{Hb}. \quad (3)$$

Subsequently, StO_2 is derived using equation (4),

$$StO_2 = \frac{C_{HbO_2}}{C_{HbO_2} + C_{Hb}} \times 100. \quad (4)$$

We implemented this data analysis method in accordance with the algorithm reported previously [22, 43]. The MATLAB-based data processing scripts used in this study are included in the supplementary material.

2.3. Device characterization

2.3.1. Configuration-dependent noise assessment

As mentioned earlier, the MAXM86146EVSYS module offers configurable LED driving settings and data acquisition parameters, including forward current amplitude, pulse width, and sampling rate. To determine the optimal configuration, we evaluated the signal-to-noise ratio (SNR) of the light intensity detected over different settings. Measurements were performed by positioning the probe on a tissue-mimicking solid phantom (Part No. 75020, ISS, USA) under consistent ambient room lighting. We first fixed the pulse width and sampling rate at 117.3 μs and 400 sps with 16 averages; then varied the LED current between 10 to 120 mA to determine the highest SNR for each wavelength. The forward current setting providing the highest SNR was then used in subsequent experiments, in which the pulse width was varied across four predefined values (14.8, 29.4, 58.7, or 117.3 μs). For each pulse width, five different sampling rates, ranging from 20 to 400 sps, were tested to assess their influence on the signal. One minute of continuous data was recorded at each setting. Then, the temporal average ($\langle I \rangle$) and standard deviation (σ) of the detected intensity were computed to estimate the SNR, using:

$$\text{SNR} = 10 \log \left(\frac{\langle I \rangle}{\sigma} \right) [\text{dB}]. \quad (5)$$

Although the MAXM86146EVSYS includes on-chip ambient light subtraction, residual signal fluctuations were still observed. These fluctuations likely originate from instrumental noise, including residual electronic noise, ADC quantization error, photodetector shot noise, and minor ambient interference that may not be fully canceled. Note that higher SNR values indicate better signal consistency over time and lower influence of internal noise on the signal.

2.3.2. Stability

To assess the stability of the instrument, continuous recordings of light intensity were acquired over 230 min, beginning immediately after powering on the system. The NIRS device was securely mounted on the same tissue-mimicking phantom used in section 2.3.1. To capture potential warm-up effects, the device remained powered off for at least 10 h prior to this test. The system was operated at the optimized settings described above, including a 400 sps acquisition rate with 16-sample averaging, 58.7 μs LED pulse width, and 120 mA driving current. The battery voltage was monitored throughout the measurement to evaluate its potential impact on LED illumination power. This was recorded using a home-built ADC, implemented with an Arduino Uno and sampled at 1 Hz.

In addition to monitoring the stability of the raw intensity, the consistency of μ_a estimation over time was assessed using the dual-slope method. This analysis required prior knowledge of μ'_s at each wavelength. According to the manufacturer, the phantom optical properties at 690 nm were $\mu'_s = 10.6 \text{ cm}^{-1}$ and $\mu_a = 0.103 \text{ cm}^{-1}$, and at 830 nm were $\mu'_s = 9.6 \text{ cm}^{-1}$ and $\mu_a = 0.101 \text{ cm}^{-1}$. A wavelength-dependent power-law model was then used to estimate μ'_s at the wavelengths of interest [7, 44, 45], yielding values of 10.2 cm^{-1} and 9.5 cm^{-1} at 735 nm and 850 nm, respectively. These values were used in the dual-slope analysis to compute μ_a over the measurement period.

2.4. Depth sensitivity assessment using layered blood phantoms

To assess the depth sensitivity of the NIRS device, we conducted a series of measurements using two-layer blood-yeast phantoms. A custom-designed hexagonal container, with side lengths of 17.5 cm and a height of 10 cm, was used (figures 3(A) and (B)). Each of the six side panels featured a transparent window sealed with a 10 μm thick Mylar sheet to accommodate optical device placement while preventing fluid leakage. To enable layered phantom measurements, an additional Mylar sheet was positioned behind five of the windows at defined distances from the outer surface. This configuration created superficial-layer compartments with thicknesses ranging from 5 to 15 mm in 2.5 mm increments. The sixth window was left without an internal partition to provide direct access to the central solution, enabling its use as a homogeneous reference medium. The base of the phantom container included an aluminum sheet to provide thermal contact with a hot plate. A detailed description of the phantom-container design is provided in the supplementary material. To assess the impact of superficial layer thickness, three identical copies of the device were utilized simultaneously. One module was placed over the window without the internal partition (i.e. directly in contact with the central solution), serving as a control. The remaining two were positioned over panels featuring a top-layer thickness of 5 mm and 15 mm, respectively, to mimic the typical scalp and skull thicknesses in neonates and adults. All modules were placed at the center of the windows to minimize signal contamination from photon reflection at the edges of the container.

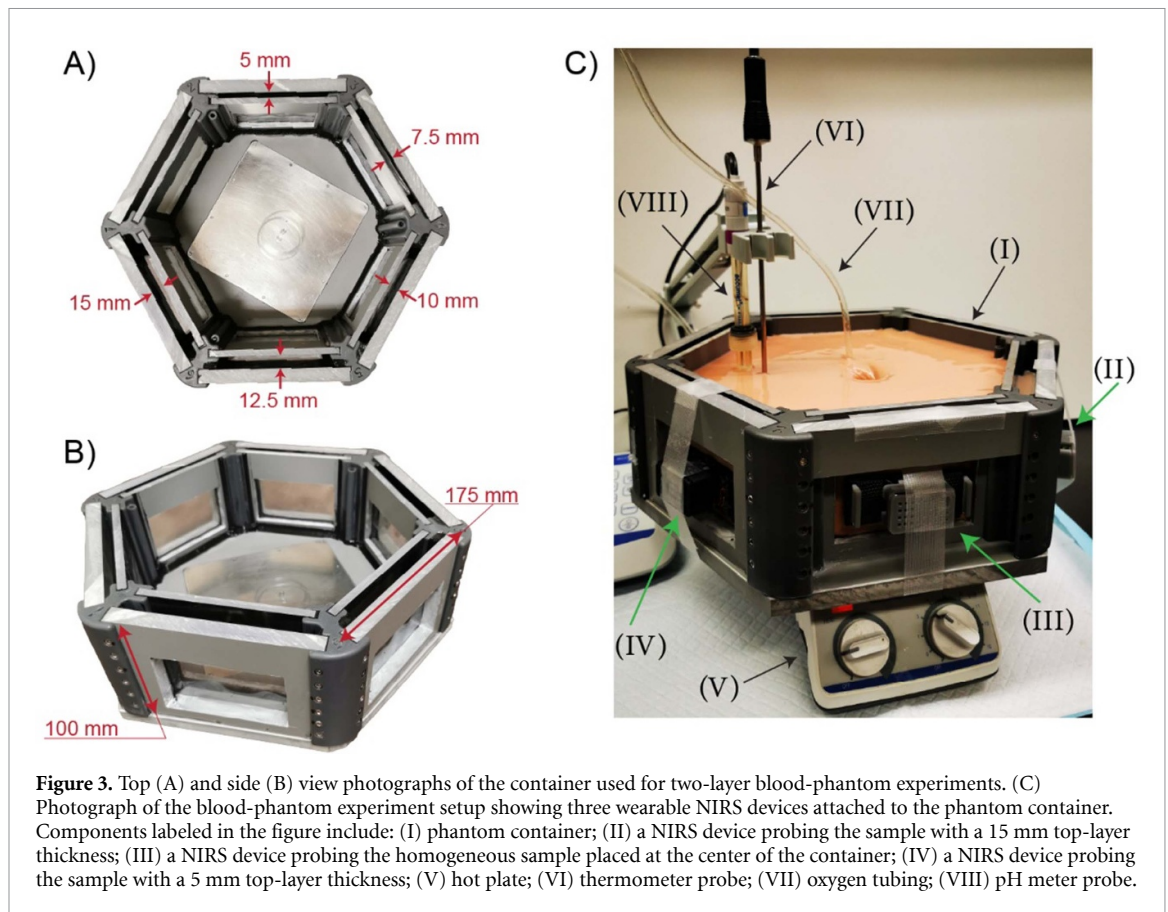


Figure 3. Top (A) and side (B) view photographs of the container used for two-layer blood-phantom experiments. (C) Photograph of the blood-phantom experiment setup showing three wearable NIRS devices attached to the phantom container. Components labeled in the figure include: (I) phantom container; (II) a NIRS device probing the sample with a 15 mm top-layer thickness; (III) a NIRS device probing the homogeneous sample placed at the center of the container; (IV) a NIRS device probing the sample with a 5 mm top-layer thickness; (V) hot plate; (VI) thermometer probe; (VII) oxygen tubing; (VIII) pH meter probe.

The phantom composition was similar to that of previous reports [32, 46–51]. Briefly, a 4.5 l solution was prepared by mixing 50 ml of whole animal blood into a mixture of Intralipid-20% (as scatterer), water, phosphate-buffered saline, and sodium-bicarbonate buffer (SBB). To mimic the light scattering of human tissue, the Intralipid-20% concentration was set to 5% providing $\mu'_s(735\text{ nm}) = 11\text{ cm}^{-1}$ and $\mu'_s(850\text{ nm}) = 9.5\text{ cm}^{-1}$ [41, 52]. All compartments were filled with the same solution to ensure consistent optical properties between layers. The container was placed on a hot plate to maintain the phantom solution at a constant temperature of 37 °C throughout the experiments. The temperature and pH of the sample were monitored continuously using a thermometer and a pH meter. A photograph of the blood-phantom experiment is shown in figure 3(C).

The measurement started by recording a 9 min baseline. Following baseline measurements, 4 g of baker's yeast was added to the phantom in the central compartment to start the deoxygenation process. The yeast was pre-activated in a solution of 4 ml of 50% glucose and 1 ml of SBB. The phantom was deoxygenated for 25 mins, and then oxygen gas was introduced into the bottom layer (i.e. the central compartment) via an air tank to reoxygenate the solution. The top layers' oxygenation was not altered throughout the experiment (i.e. remained fully oxygenated at all stages), thereby allowing for assessment of the device's sensitivity to deep-layer oxygenation changes under various superficial layer thicknesses.

2.5. Evaluation across diverse skin tones

To evaluate the system's performance across different skin pigmentation levels, we performed a series of phantom and *in vivo* measurements. The phantom measurement consisted of placing thin, pigmented layers on a homogeneous blood phantom to assess the effect of skin pigmentation on StO₂ estimation. For this purpose, we constructed a homogeneous blood phantom following the protocol described in section 2.4 and used the same container illustrated in figure 3(B). In this experiment, only the window providing direct access to the central solution of the phantom (homogeneous sample) was utilized. To simulate different skin pigmentation levels, we used a set of four thin (1 mm-thick) tissue-mimicking solid phantoms corresponding to Fitzpatrick skin phototypes II, III, V, and VI (figure 4), as reported in [53].

We began with the lightest phantom layer (Fitzpatrick II), placing it directly on the exterior surface of the Mylar sheet covering the chamber window. The NIRS probe was positioned on top of this solid phantom layer, and a 90 s recording was performed. The solid phantom layer was then sequentially

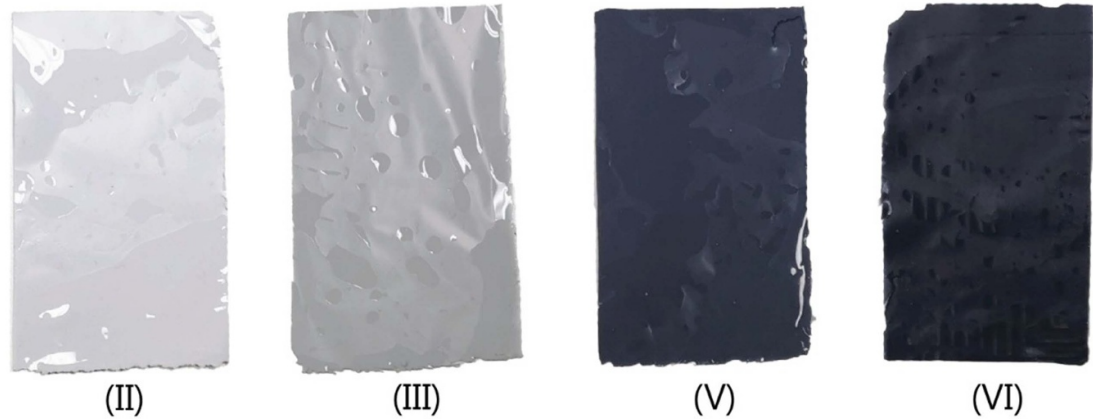


Figure 4. Photograph of the 1 mm-thick tissue-mimicking solid phantoms used to simulate different skin pigmentation levels, corresponding to Fitzpatrick skin phototypes II, III, V, and VI.

replaced with increasingly darker skin phantoms, and for each pigmentation level a 90 s data acquisition was performed. To enable direct comparison and quantify the effect of skin pigmentation on StO_2 estimation, a commercial trNIRS device (NIRSBOX-Q, PIONIRS, Italy) was used to acquire trNIRS data before replacing the phantom layer at each step of the experiment. The trNIRS device operated at an acquisition rate of 0.5 Hz; therefore, the wearable NIRS signal was down-sampled to 0.5 Hz (by averaging over 2 s intervals) to match the trNIRS temporal resolution for direct comparison. During all recordings, the blood phantom oxygenation level was kept constant without adding any yeast or inducing deoxygenation.

In vivo measurements were acquired on three adult volunteers with skin tones visually spanning a gradient from Fitzpatrick type II to type VI. The optical device was secured to the forehead using a self-adhesive bandage, and a 10 s recording was acquired while each subject remained seated at rest. All measurements were performed using the same acquisition settings described in section 2.3.2. Detected signal intensity and HbO_2 variations, computed using single-channel analysis, were used for this assessment.

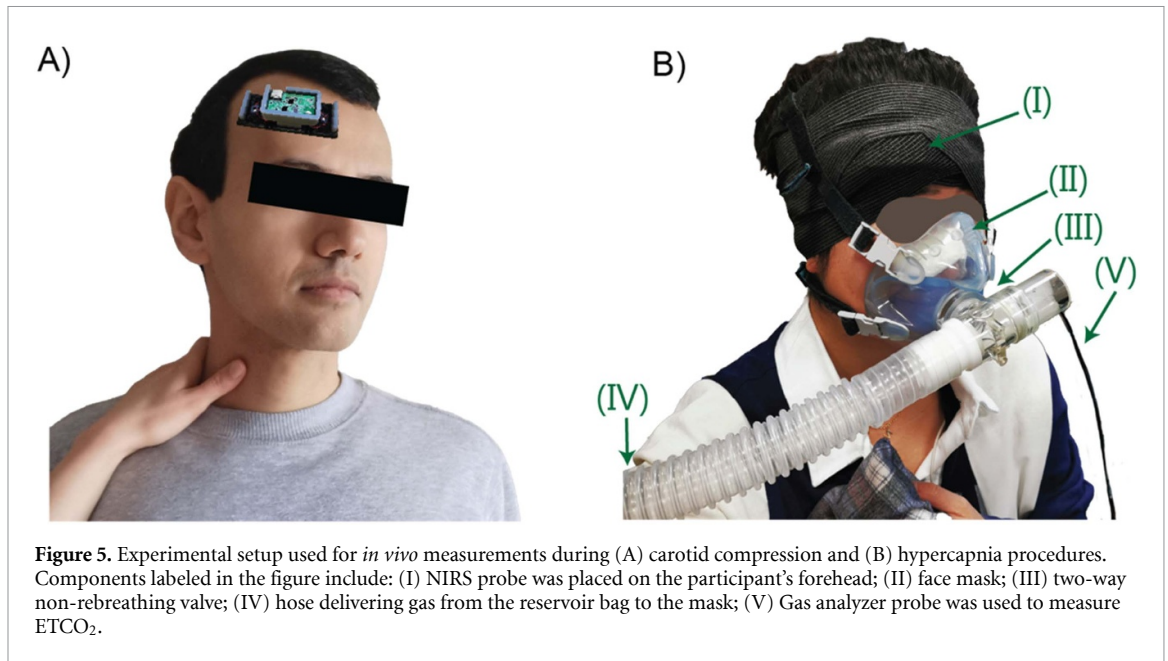
2.6. Cerebral hemodynamics measurements

To evaluate the capability of the system to monitor cerebral hemodynamics, we conducted two series of *in vivo* experiments: carotid compression (CC) and hypercapnia challenge. Each experiment included four healthy adult volunteers (3 males, 1 female; age range of 26–34). Measurements were acquired with subjects seated comfortably, and the probe secured on the right side of the forehead with a self-adhesive bandage to minimize motion artifacts.

The CC experiment began with a 1 min baseline recording to measure a stable reference signal. Then, the right common carotid artery (i.e. ipsilateral to the NIRS probe) was fully compressed for 15 s, as illustrated in figure 5(A). Compressing the carotid artery reduces cerebral blood flow from the ipsilateral side. After the compression, a 1 min recovery period was recorded, during which normal circulation resumed. The compression was performed manually by trained study personnel. During the last 10 s of the baseline period, the artery was located by palpation in the lower portion of the neck, where the carotid pulse is easily felt. Throughout the 15 s occlusion, firm and steady pressure was applied with the personnel's fingers to block the blood flow.

Hypercapnia was induced using the Douglas bag method. Participants wore a sealed face mask connected to a two-way non-rebreathing valve and a reservoir bag containing 5% CO_2 and 95% oxygen concentration (figure 5(B)). The protocol consisted of a 1 min baseline (room air), 5 min hypercapnic exposure, and 2 min recovery. Partial pressure of end-tidal CO_2 (ETCO_2) was continuously monitored (ML206 Gas Analyzer, ADInstruments) to confirm elevated arterial CO_2 levels. Hypercapnia increases arterial CO_2 tension, leading to cerebral vasodilation and elevated cerebral blood flow [54]. The hemodynamic response to this stimulus is expected to involve an increase in HbO_2 and either a stable or slightly decreased Hb [55].

To calculate StO_2 time courses using the dual-slope approach, the μ'_s of the probed tissue is required. As described in section 2.2, these values can be obtained from literature or measured using quantitative optical techniques. Here, we determined the μ'_s using a trNIRS system (NIRSBOX-Q, PIONIRS,



Italy) to improve the accuracy of our analysis. For each participant, μ'_s values were obtained with the trNIRS system at four wavelengths: 760, 808, 850, and 915 nm. The probe was positioned at the same forehead location as the wearable NIRS device. The μ'_s value at 805 nm was obtained directly from the time-resolved measurements, and the μ'_s at 735 nm was estimated using the wavelength-dependent power-law model fitted to the μ'_s values measured at the four time-resolved wavelengths [7]. The μ'_s values measured across all participants were averaged ($\mu'_s(735 \text{ nm}) = 12.9 \pm 0.8 \text{ cm}^{-1}$ and $\mu'_s(850 \text{ nm}) = 10.4 \pm 0.5 \text{ cm}^{-1}$) and used to compute StO_2 variations.

For both the CC and hypercapnia measurements, maximum changes in HbO_2 , Hb , and StO_2 were compared to the baseline. Given the proof-of-concept nature of this study, which has a small sample size ($n = 4$) unsuitable for normality testing, results are reported in terms of effect sizes (Cohen's D) and 95% confidence intervals (CI) rather than p -values. Subsequently, precision-based sample size estimation was conducted to determine the number of participants required to achieve a 95% CI with a maximum width of $\pm 1\%$ for StO_2 .

3. Results

3.1. Signal quality assessment

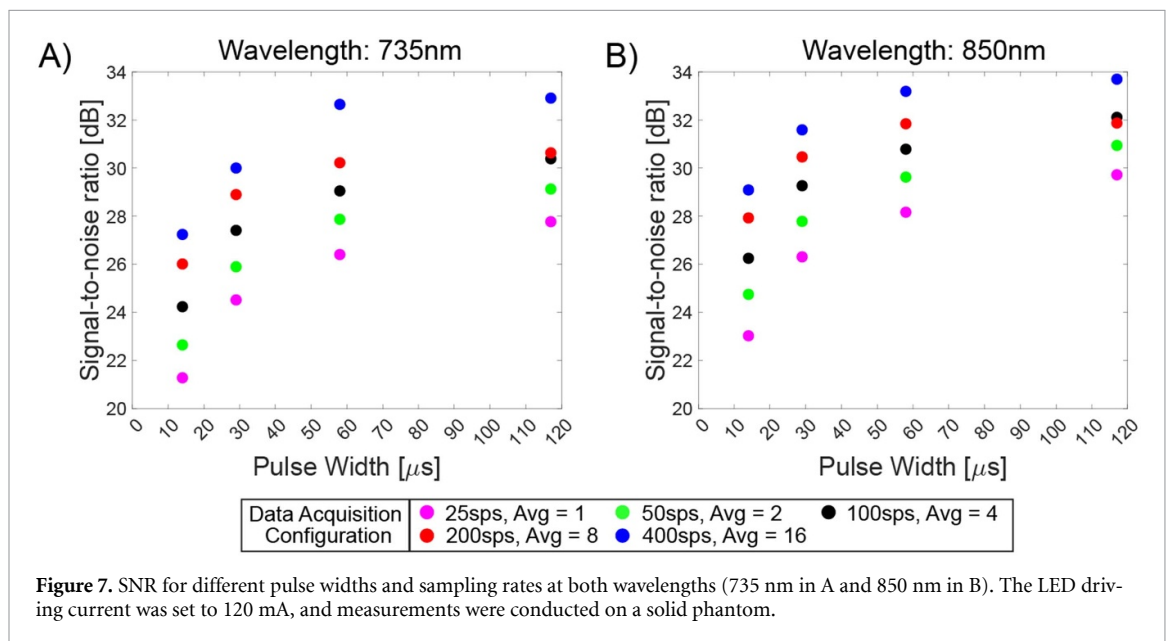
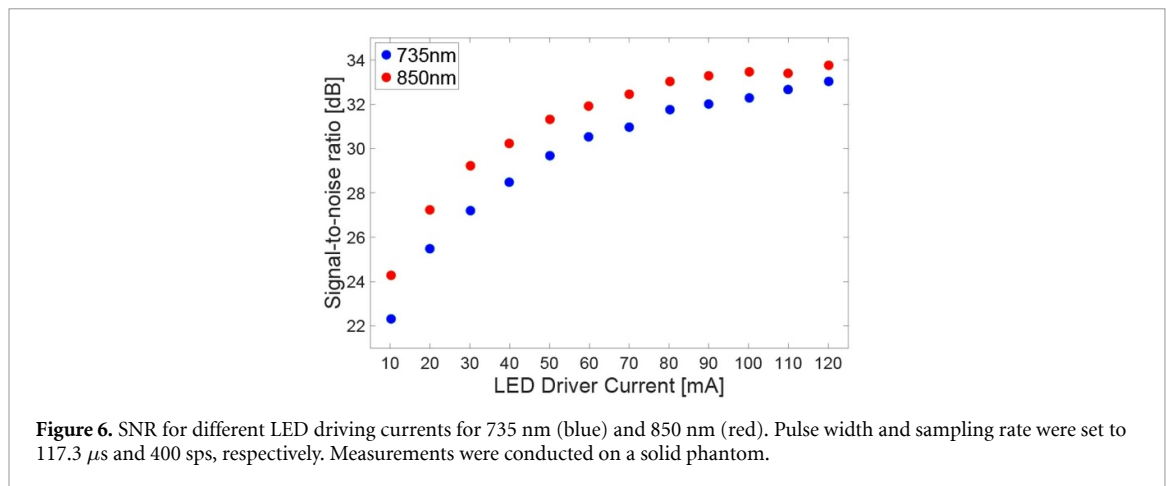
Figure 6 shows the SNR as a function of LED forward current for the two wavelengths. In both cases, SNR increased with current, reflecting the higher optical power delivered to the detectors due to higher illumination intensity. The maximum SNR for both wavelengths was achieved when the LEDs were driven at 120 mA, which is the maximum current supported by the module's built-in drivers.

Figure 7 shows the impact of pulse width and sampling rate on signal quality. Both wavelengths showed similar behavior, with the SNR increasing for longer pulse widths and higher sampling rates. At each pulse width, the highest SNR was obtained at a sampling rate of 400 sps, which, after averaging every 16 samples, yielded an effective output rate of 25 Hz, as described in the [Methods](#) section.

Although the highest overall performance was measured at a pulse width of 117.3 μs and a sampling rate of 400 sps, the SNR difference between 117.3 μs and 58.7 μs pulse widths was relatively minor ($< 0.5 \text{ dB}$). Considering the trade-off between signal quality and power consumption, a pulse width of 58.7 μs was selected as the optimal configuration, providing near-maximal SNR while reducing energy consumption. Hence, the optimal settings of the device included an LED current of 120 mA, a pulse width of 58.7 μs , and a sampling rate of 400 sps with averaging over 16 samples.

3.2. Stability

Figure 8(A) shows the mean normalized light intensity at 735 nm (blue) and 850 nm (red) over a 230 min measurement period immediately after powering on the NIRS module. For each wavelength, the intensity curves from the four source-detector pairs were first normalized to the mean value during the last 10 min and then averaged. A small initial decline was observed after system power-up; however, the



temporal variation remained below 1% throughout the measurement, demonstrating the high stability of the system. To evaluate the potential impact of this brief warm-up period on the performance of the system, μ_a was computed using the dual-slope approach. As shown in figure 8(B) the recovered μ_a values were close to the control (0.1 cm^{-1} , as reported by the manufacturer) and remained stable for the 230 min period for both wavelengths, with a negligible drift. Monitoring the battery voltage (figure 8) showed an initial value of 4.07 V, which decreased by 10% after 190 min to 3.66 V. Continuous operation for 230 min reduced the voltage to 3.63 V, corresponding to 89.2% of the initial value.

3.3. Depth sensitivity assessment using a two-layer blood phantom

Figure 9 displays the StO_2 time courses obtained during the blood phantom experiments. Simultaneous recording was conducted using three duplicates of the NIRS device and analyzed based on the dual-slope approach to assess depth-sensitivity. Measurements acquired through 5 mm (blue line) and 15 mm (black line) thick top layers were compared to control values obtained directly from the deep-layer compartment (red line).

The same solution was used in both compartments at the beginning of the experiment; therefore, both layers had the same oxygenation level at the start of the measurement. As a result, all three StO_2 curves begin at a very similar baseline value of $72 \pm 0.12\%$. After initiating the deoxygenation process by adding yeast to the bottom compartment, oxygen was gradually consumed, and the StO_2 magnitude dropped to $46 \pm 0.08\%$. The StO_2 obtained through the 5 mm static superficial layer (blue curve) closely matches the control in both amplitude ($47 \pm 0.06\%$) and temporal dynamics. In contrast, the compartment with a 15 mm top-layer thickness (black curve) shows less StO_2 changes ($60 \pm 0.06\%$), indicating attenuation of the signal deep-sensitivity due to the thicker overlying layer. Nevertheless, despite the

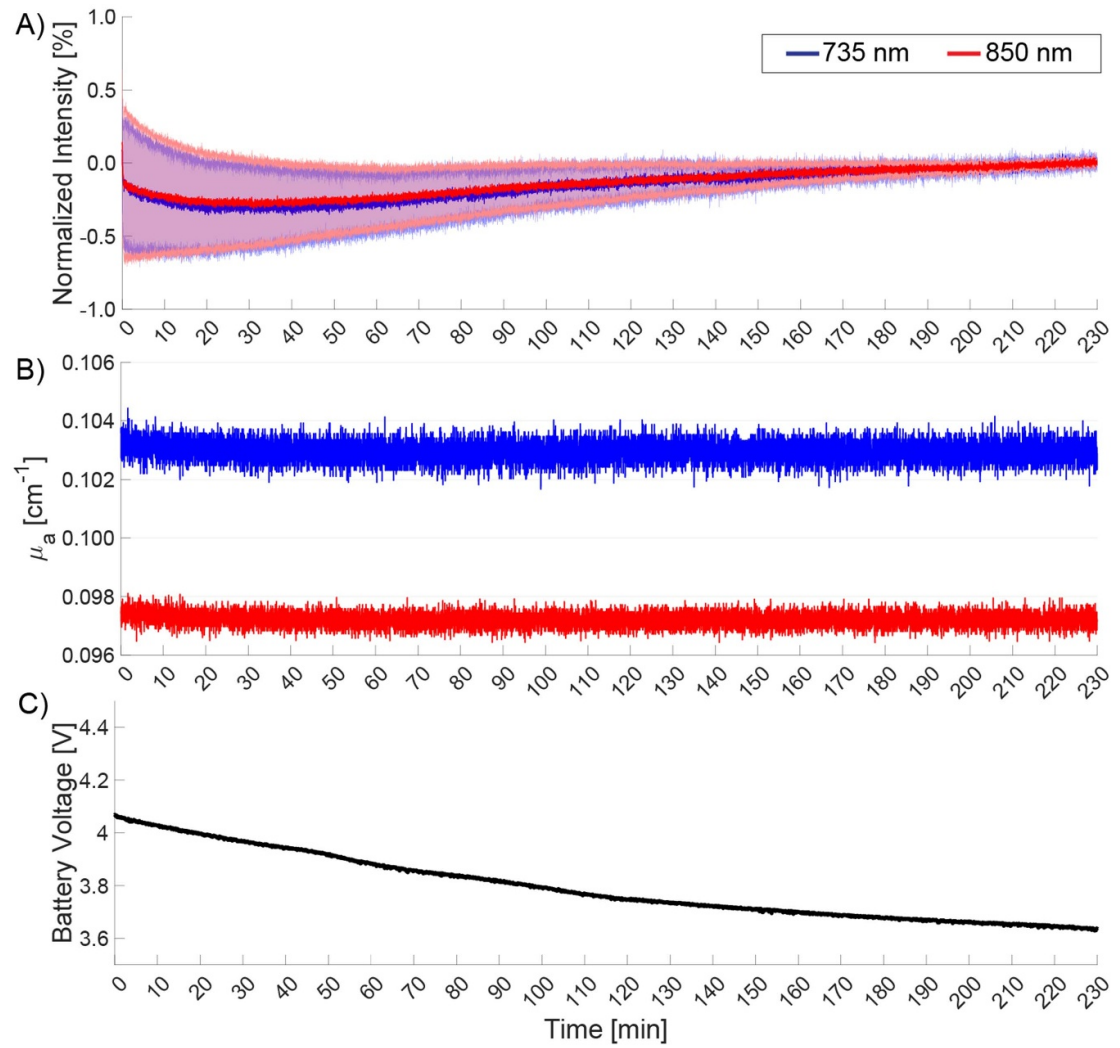


Figure 8. System stability assessed by monitoring normalized light intensity (A) and absorption coefficient consistency (B). (C) Battery Voltage. The detected light intensities in panel (A) were recorded continuously over 230 min and normalized to the mean value of the final 10 min. The solid lines indicate the average across four source-detector pairs, and the shaded areas represent the standard deviation. In each panel, results at 735 nm and 850 nm are shown in blue and red, respectively.

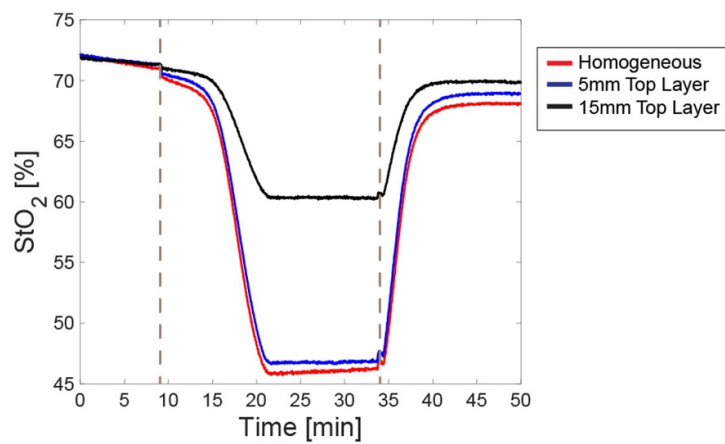
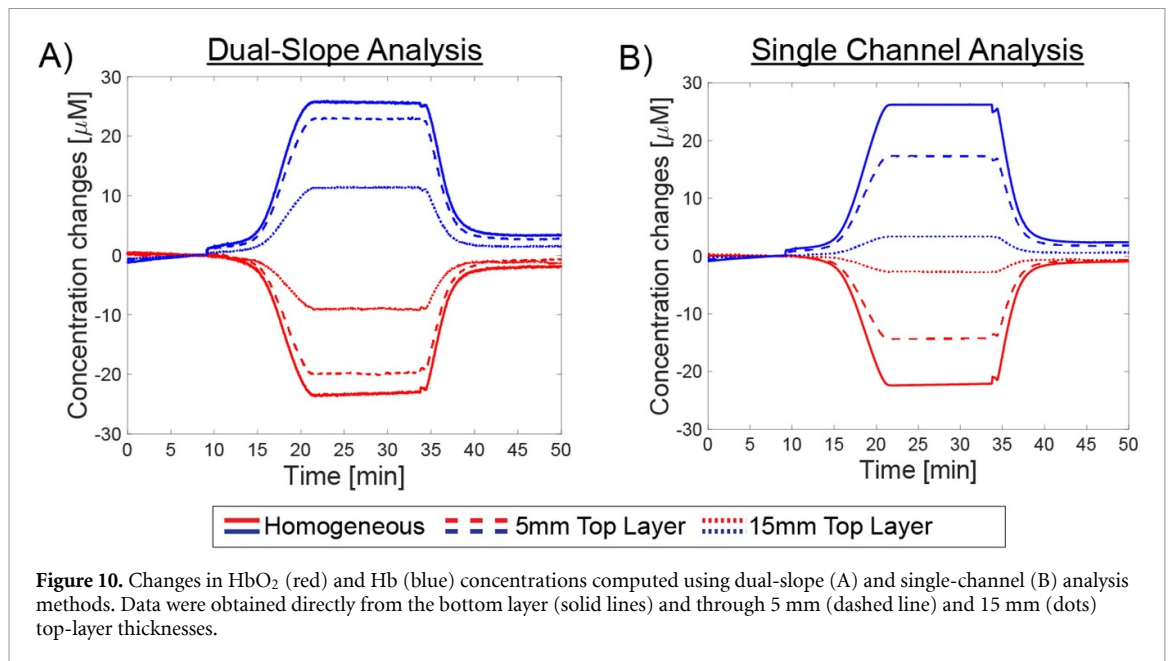


Figure 9. Simultaneous measurements of StO₂ using three duplicates of the NIRS device on two-layered blood phantoms. The red curve represents the control measurement recorded from the homogeneous bottom layer. Blue and black curves correspond to measurements with 5 mm and 15 mm top-layer thickness, respectively. StO₂ values were computed based on the dual-slope analysis.



underestimation in absolute values, the system was able to detect oxygenation variations in the deep layer, demonstrating its capability to detect changes in cerebral oxygenation beneath superficial layers typical of adults.

Figure 10 compares changes in hemoglobin concentrations (HbO₂ in red, Hb in blue), using the same dataset shown previously, but processed with the two analysis approaches to assess their depth sensitivity. Figure 10(A) shows results obtained using the dual-slope method, while the outputs from the conventional single-channel method, based on the MBLL, are displayed in figure 10(B). In both cases, the measurements were acquired from three configurations: directly from the bottom layer (solid lines), and through 5 mm and 15 mm top layers, indicated by dashed lines and dots, respectively. Please note that for each phantom, the total hemoglobin concentration remained constant, resulting in inverse changes between Hb and HbO₂. Thus, an increase in Hb corresponds to proportional decreases in HbO₂. As expected, both methods yielded comparable results for the control sample, where light only travelled through the homogeneous deep layer. Table 2 shows that when a superficial layer was present, the single-channel analysis underestimated the magnitude of hemodynamic changes occurring in the bottom layer more than the dual-slope analysis. Notably, the dual-slope approach recovered these changes with higher precision, demonstrating enhanced depth sensitivity. These findings underscore the limitations of conventional analysis in layered media and highlight the advantage of using depth-resolved approaches for deep tissue monitoring.

3.4. Evaluation across diverse skin tones

Figure 11 compares the StO₂ values of the homogeneous blood phantom under different simulated skin pigmentations, achieved by covering the sample with thin solid phantoms. The measurements obtained with the wearable NIRS device and the trNIRS system are represented by the red and blue bars, respectively. The trNIRS system recovered a consistent StO₂ of $80.7 \pm 0.3\%$ for all cases, as expected from the homogeneous and oxygenated phantom. In contrast, the StO₂ estimated by the wearable NIRS device showed a gradual upward shift with increasing pigmentation. The recovered value was close to the trNIRS reference for the lightest skin phantom (Fitzpatrick II), with StO₂ = $79.5 \pm 0.1\%$. However, the measurement accuracy decreased with darker pigmentation levels, resulting in an overestimation of 10% for Fitzpatrick VI (StO₂ = $89.5 \pm 0.8\%$).

As indicated in figure 12, the amplitude of the detected light intensity varied across participants. For the light-skinned subject (figure 12, top row), the raw signal had the highest amplitude and exhibited a clear pulsatile waveform. In contrast, increasing skin pigmentation resulted in reduced signal amplitude and diminished pulsatility, with the darkest-skinned subject (figure 12 bottom row) showing the lowest visibility of the pulsations in the raw intensity trace.

Despite this attenuation, changes in hemoglobin oxygenation derived using single-channel analysis revealed clear pulsatile waveforms in all three participants. This outcome is attributed to the system's built-in ambient light cancellation and low-noise AFE, which preserves signal fidelity even under

Table 2. Comparison of dual-slope and single-channel analyzes in recovering bottom-layer hemodynamic changes under different superficial layer thicknesses.

	Dual-slope analysis					Single channel analysis				
	Bottom layer (Control) (μM)	Top layer thickness				Bottom layer (Control) (μM)	Top layer thickness			
		5 mm		15 mm			5 mm		15 mm	
		Value (μM)	Error [%]	Value (μM)	Error (%)		Value (μM)	Error (%)	Value (μM)	Error (%)
ΔHbO_2	−23.3	−19.9	14.6	−9.0	61.4	−22.3	−14.3	35.9	−2.7	87.9
ΔHb	25.7	22.9	10.9	11.4	55.6	26.2	17.3	34.0	3.4	87.0

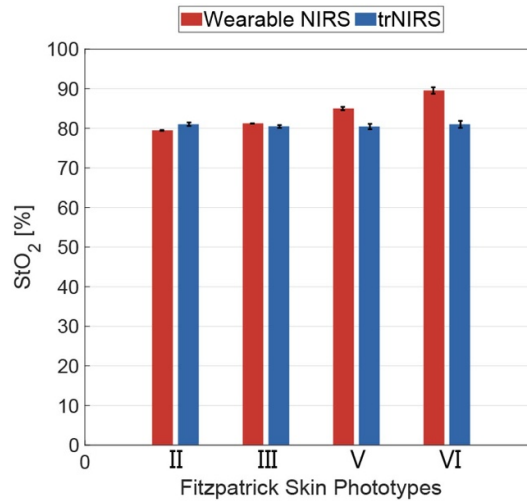


Figure 11. StO₂ values measured by the wearable NIRS device and the trNIRS system on a homogeneous blood phantom covered by thin solid phantoms representing different skin tones. The error bars represent the standard deviation from 45 datapoints.

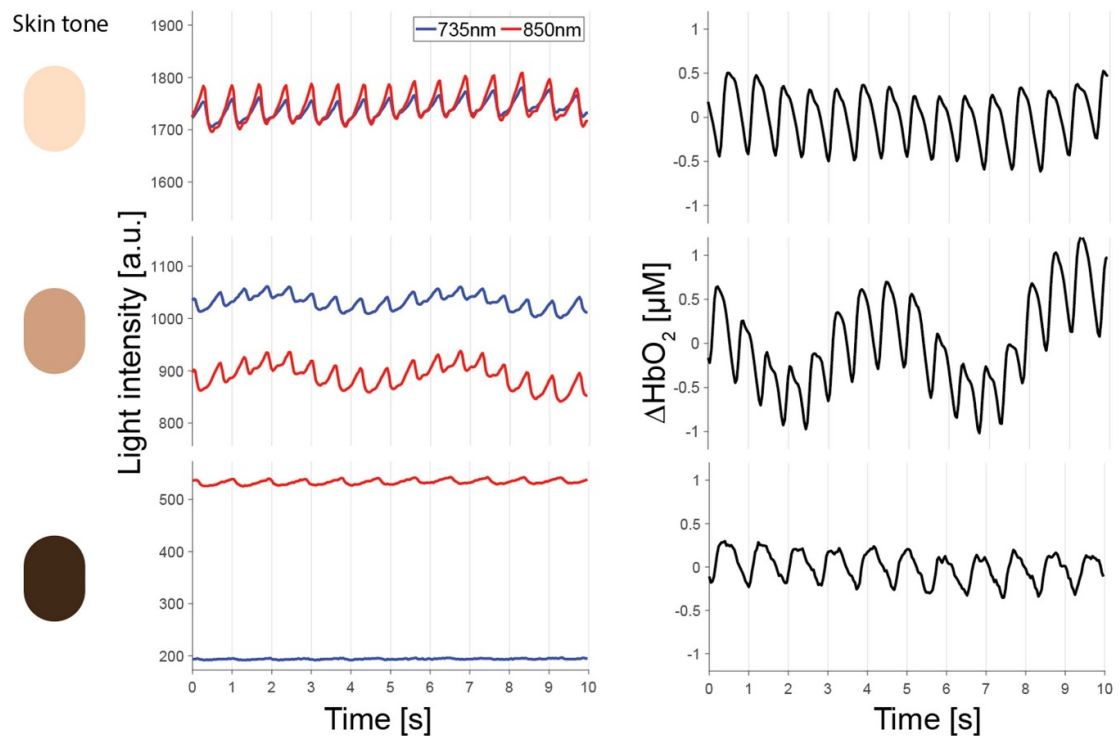
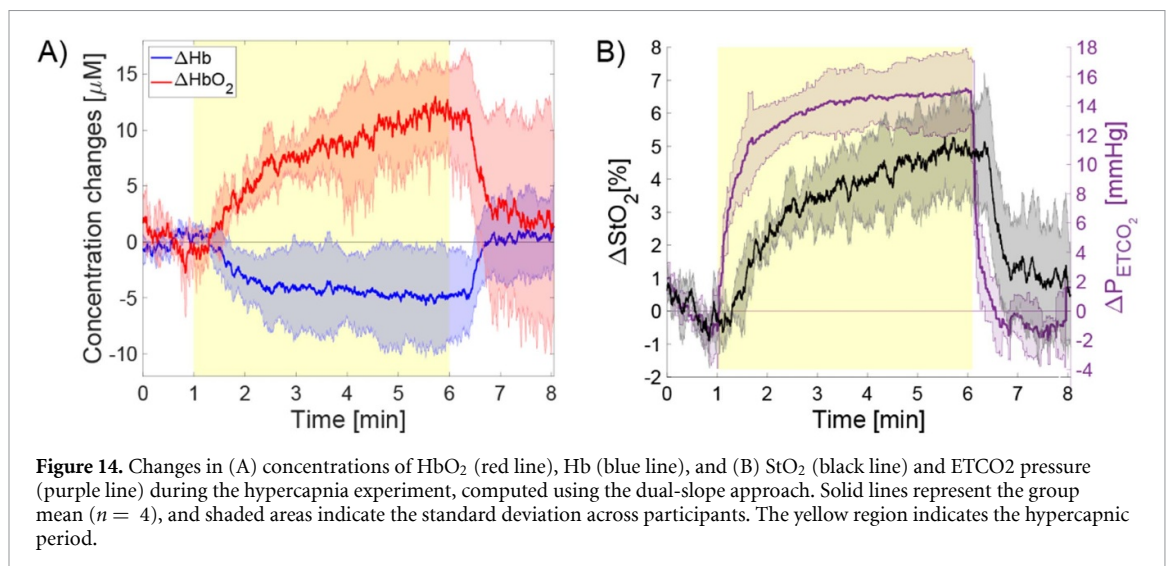
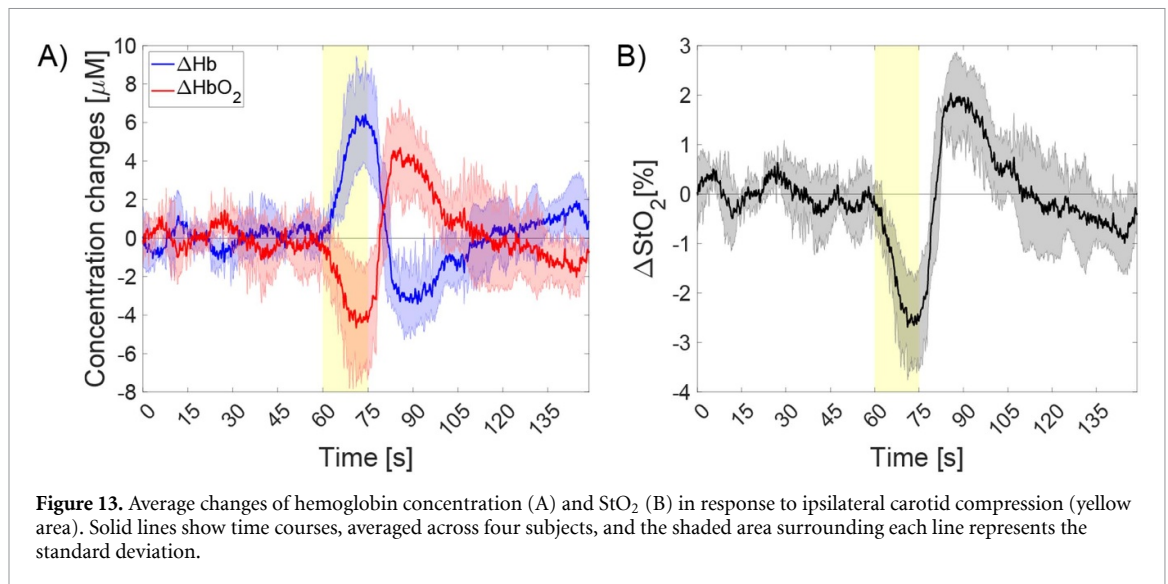


Figure 12. Measurements were conducted on three volunteers with various skin tones (left column). The detected light intensity (middle) and ΔHbO_2 (right) signals were recorded over 10 s. All measurements were acquired on the forehead using identical acquisition settings and processed using single-channel analysis.

low-intensity conditions. As shown in figure 12, right column, heart-rate-modulated variations in HbO₂ are evident for all skin tones. These findings confirm that while melanin reduces the detected intensity, the device's signal processing pipeline can recover physiologically meaningful hemodynamic signals across a broad range of pigmentation levels. Due to differences in detected light intensity across subjects, each graph in the middle column of figure 12 uses a distinct y -axis range; however, the dynamic range remains consistent, allowing valid comparison of pulsatility.

3.5. In vivo experiments

The results of the CC experiment (figure 13) illustrate the system's ability to detect physiologically relevant changes in cerebral tissue hemodynamics and oxygenation in response to transient vascular manipulation. A drop in HbO₂ concentration (figure 13(A)) (Cohen's $D = 2.1$; 95% CI = -8.3 to



0.3 μM), along with an increase in Hb concentration (figure 13(A)) (Cohen's $D = 3.6$; 95% CI = 2.3–9.6 μM) and a decrease in StO₂ (figure 13(B)) (Cohen's $D = 3.8$; 95% CI = –3.9 to –1.0%), were observed upon initiation of compression, reflecting reduced oxygen delivery to the ipsilateral hemisphere. The mean reduction in StO₂ was $2.5 \pm 0.91\%$, which is consistent with previous findings using hyperspectral NIRS and trNIRS [54, 56]. The cerebral hemodynamic variations started 2.5 s after the beginning of the carotid occlusion, when HbO₂ and Hb variation became larger than baseline hemodynamic fluctuations. An estimated sample size of 6 participants is required to achieve a precision of $\pm 1\%$ for the 95% CI of StO₂. Following the release of the compression, a hyperemic response was observed before StO₂ returned to baseline. This expected hemodynamic pattern confirms the system's ability to detect changes in cerebral hemodynamics associated with vascular occlusion.

The results of the hypercapnia experiment are presented in figure 14, showing the measured changes in hemoglobin concentrations (figure 14(A)) and the corresponding StO₂ variation over time (figure 14(B)). During the 5 min hypercapnic period, HbO₂ concentration increased by $12 \pm 3.8 \mu\text{M}$ (Cohen's $D = 4.6$; 95% CI = 6.0–17.4 μM), leading to a rise in StO₂ of $4.9 \pm 1.5\%$ (Cohen's $D = 5.0$; 95% CI = 2.7%–7.1%). In contrast, the mean change of $-4.9 \pm 4.5 \mu\text{M}$ in Hb had a lower effect (Cohen's $D = 1.6$; 95% CI = –11.7–2.0 μM), suggesting that Hb remained unchanged or even increased in some participants, as expected [55, 57]. An estimated sample size of 10 participants is required to achieve a precision of $\pm 1\%$ for the 95% CI of StO₂. Monitoring ETco₂ pressure revealed an average increase of $15 \pm 2.7 \text{ mmHg}$ in the partial pressure of exhaled CO₂ during hypercapnia, with StO₂ showing a corresponding increase that followed the ETco₂ dynamics (figure 14(B)). After CO₂ inhalation ceased, both hemoglobin concentrations and StO₂ returned to baseline during the recovery phase.

4. Discussion

To address the need for accessible neonatal brain monitoring tools in resource-limited countries, we repurposed an optical biosensing module to develop a low-cost, wearable cerebral oximeter. This approach eliminates the need for complex custom hardware development, simplifies data acquisition, and requires only basic engineering skills to build a wearable NIRS device, making it practical for low-resource settings. The system satisfies the essential requirements for a cerebral oximeter in underserved regions, particularly for neonatal applications. Its characteristics include:

- Compact, lightweight design for comfortable long-term wear
- Wireless data transmission for wearability
- Rechargeable power supply, supporting up to 4 h of continuous monitoring
- Reliable tissue oxygenation measurements without the need for calibration
- User-friendly interface with real-time raw data (i.e. detected light intensity) streaming
- High SNR from integrated noise-cancellation, enabling reliable measurements across different skin tones, and operation under ambient lighting
- Rapid signal stabilization upon power-up, facilitating plug-and-play operation without warm-up requirements

The system has a customizable architecture that allows for flexible configurations to suit different applications. Further, its straightforward assembly process and low maintenance enable non-experts to assemble, operate, and modify the device with only minimal training or technical background.

To validate the feasibility of the proposed wearable NIRS system, we conducted a series of evaluations, including system performance tests, phantom measurements, and in vivo experiments. Since reliable brain monitoring requires stable measurements with high SNR, we first assessed the SNR under different settings. The results presented in figures 6 and 7 demonstrate that increasing the LED driving current, pulse width, and sampling rate enhances SNR. This improvement is associated with higher optical power reaching the detector, thereby increasing the signal amplitude relative to the noise level, which is driven by ambient light and internal sources such as electronic noise, shot noise, and ADC quantization. To optimize the trade-off between SNR and power consumption, the configuration included an LED current of 120 mA, a pulse width of 58.7 μ s, and a sampling rate of 400 sps. These settings provide near-maximal SNR while reducing energy demand compared to the longest pulse width. We also evaluated the impact of driving the LEDs at 120 mA during prolonged operation by estimating the corresponding junction temperature. The analysis indicated that the thermal rise remains within safe limits without affecting LED performance, as detailed in the supplementary material.

The system's performance was also assessed by monitoring the temporal variation of the detected light intensity over 230 min. This test was conducted by positioning the probes on a tissue-mimicking solid phantom and powering on the system. The results presented in figure 8 provide a quantitative assessment of the system stability. The recorded light intensity exhibited less than 1% variation across all source-detector pairs, indicating that the system maintains a high degree of temporal stability. Unlike many NIRS systems, particularly trNIRS devices, that require a warm-up period of at least 20–60 min [56, 58–60], our device can be operated immediately after powering on, with no need for an extended warm-up period. While an initial reduction in detected intensity was observed, its magnitude was negligible across all channels and did not meaningfully affect data quality. This is further supported by the μ_a estimates derived using the dual-slope method, which remained stable throughout the measurement (figure 8(B)).

We next assessed the system's depth sensitivity using layered blood phantoms, as described in section 2.4. Figure 9 demonstrates that the developed NIRS system is well-suited for neonatal brain monitoring, since measurements in the presence of a thin superficial layer (3–5 mm) [61, 62] closely followed the reference measurements acquired without any superficial layers. In addition, the results from the sample with a 15 mm superficial layer demonstrate the system's potential for trend monitoring of adult cerebral hemodynamics [57, 63]. Although the magnitude of deep-layer oxygenation changes was underestimated, the system successfully detected the trend of the hemodynamic changes. This capability is clinically valuable and consistent with functional NIRS (fNIRS) applications, as relative changes in cerebral oxygenation can serve as early indicators of impaired brain perfusion and cognitive dysfunction [64–67].

To further investigate depth sensitivity, we compared dual-slope and single-channel data processing approaches. As noted in the Method section, the source-detector arrangement of the system allows data processing using either the conventional single-channel approach or the dual-slope method. To validate

the enhanced depth sensitivity of the dual-slope approach, we compared its performance with the single-channel method by computing hemodynamic variations from the same dataset. In an ideal system with high depth sensitivity, even in the presence of superficial layers, the recovered hemodynamic variations from the deep layer should closely match those obtained from the control medium (i.e. without superficial layers). In practice, this is effectively unachievable, particularly for cwNIRS due to its inherent depth-sensitivity limitations. Nevertheless, the results in figure 10 indicate that dual-slope analysis provides better agreement with the homogeneous reference than the single-channel method, especially when the medium is covered by a 15 mm superficial layer. These findings highlight the advantage of dual-slope analysis in reducing superficial contamination and emphasize the importance of equipping cwNIRS systems with appropriate source-detector configurations to leverage this property.

To assess the precision of the developed NIRS system in estimating StO_2 magnitude, we performed a separate measurement on a homogeneous blood phantom using a commercial trNIRS device as a reference. The results of this experiment are provided in the supplementary material. We found that the wearable system overestimated the StO_2 , when the phantom was deoxygenated. This discrepancy is likely caused by scattering changes introduced after adding yeast, as trNIRS directly measures the sample's optical properties and therefore accounts for variations in μ'_s when computing StO_2 . In contrast, the dual-slope approach in CW regime assumes a fixed μ'_s , which impacts the precision when the sample's scattering properties drift over time. Importantly, this is not expected for most in vivo experiments, where tissue oxygenation changes are primarily driven by variations in hemoglobin concentration, while tissue scattering remains relatively stable over short experimental intervals.

It is noteworthy that the system does not restrict users to the built-in LED drivers, as external drivers could be employed to further increase illumination power. The need for additional illumination power to enhance SNR is especially relevant for individuals with darker skin pigmentation, where greater superficial light absorption reduces the detected signal. To investigate this point, we assessed the impact of skin pigmentation level on StO_2 measurement accuracy using blood phantom experiments in which the underlying oxygenation level was held constant throughout all stages. Under these conditions, the estimated StO_2 is expected to remain unchanged across all measurement stages, involving skin phantoms with various pigmentation level. The results revealed that the system exhibited a systematic upward shift in StO_2 level with increasingly pigmented superficial layers, reaching an overestimation of up to 10% for Fitzpatrick type VI. In contrast, the trNIRS system reported consistent StO_2 values across all pigmentation levels, reflecting its higher depth sensitivity compared with cwNIRS. Previous studies have reported similar pigmentation-related biases in pulse oximeters, where device accuracy is affected by skin tone [68, 69]. Moreover, we recorded in vivo steady-state data from three participants with a wide range of skin tones. As shown in figure 12, increasing melanin concentration in the skin significantly attenuated the detected light intensity. Nevertheless, the module's integrated noise-cancellation algorithm makes it possible to detect pulsatility in the raw signals even at low amplitudes, and these pulsatile waveforms become more visible after processing the raw data to obtain HbO_2 concentration. This feature addresses a critical consideration for equitable clinical deployment across diverse global populations.

In vivo experiments on adult participants confirmed the system's ability to capture cerebral hemodynamic and oxygenation changes driven by physiological variation. These results also confirm its capability to detect signals originating from beneath approximately 1.5 cm of extracerebral tissue, as reported in prior MRI-based measurements of extracerebral layer thickness [57]. While the physiological interpretation of these responses is beyond the scope of this work, these findings are discussed in the context of validating the performance and robustness of the developed NIRS system. In the CC experiment (figure 13), the device detected a rapid drop in StO_2 during common carotid artery occlusion, followed by a prompt return toward baseline after release. This response, which agrees with previous NIRS-based observations [42, 44], reflects a temporary reduction in cerebral blood flow and the subsequent hyperemic response, confirming the system's sensitivity to fast, event-driven vascular changes.

During the hypercapnia experiment (figure 14), a gradual increase in HbO_2 and StO_2 was observed during CO_2 inhalation, consistent with the expected physiological response of cerebral vasodilation and increased blood flow under elevated arterial CO_2 levels [57, 70]. The return of both HbO_2 and StO_2 toward baseline during the recovery period further illustrates the device's capacity to track dynamic, reversible changes in cerebral perfusion and oxygenation.

The in vivo measurements performed on adults, together with the results of the two-layer blood-phantom experiments, support the feasibility of using this system for neonatal cerebral monitoring. Newborns have considerably thinner extracerebral layers and more translucent skin than adults, conditions that enhance optical penetration and provide higher SNR for NIRS measurements [71]. These anatomical advantages make the introduced system a practical and suitable tool for neonatal brain monitoring in LMICs. In addition, thermal characterization tests presented in the supplementary material,

along with the illumination power of the module being below established skin-exposure safety limits (section 2.1), indicate that the proposed system is a safe and practical solution for neonatal brain monitoring, particularly in LMIC.

In addition, the system enables simultaneous monitoring of hemodynamic trends and high-frequency pulsatile waveforms at 25 Hz. This capability allows users to observe slow changes in tissue oxygenation while also capturing arterial pulsatility, which may offer additional insights into vascular compliance and cerebral autoregulation [72]. Interestingly, previous work has also demonstrated the sensitivity of slow fluctuations in peripheral tissue oxygenation to sepsis-related microvascular dysfunction [73–75], a life-threatening condition heavily burdening low-resource populations [2]. Since skeletal muscle is typically overlaid by thin skin and fat layers, the current system has the potential to be employed as an accessible and rapid sepsis monitor; however, this idea needs validation with a clinical study.

This work demonstrates that the introduced NIRS system provides a reliable, wearable, and accessible solution for non-invasive monitoring of cerebral hemodynamics. Built from an evaluation kit of an optical biosensing module, the device is low-cost, easy to assemble, and suitable for both neonatal and adult use across a variety of skin tones. This greatly enhances its potential for adoption, especially in low-resource settings.

In comparison with other wearable open-source NIRS systems listed in table 1 that require custom-fabricated printed circuit boards, professional component assembly, and custom firmware, our system takes a different approach. It uses a preassembled commercial unit, allowing the NIRS device to be built with minimal modifications: replacing the LEDs and securing the board within a 3D-printed probe. This approach improves reproducibility, particularly for low-resource settings. Notably, an assembled version of the other module may be more costly than the component prices listed in table 1, which can limit accessibility or may not be affordable in low-resource environments. Although HEGDuino is the most affordable option, with a mounted version costing under US \$100, it has been used in only a few studies [15, 21], and its technical performance has undergone limited assessments. Further, most systems in table 1 (except FlexNIRS) are not compatible with the dual-slope method and operate using conventional single-source-detector configurations that provide limited depth sensitivity and are more affected by extracerebral contamination. In contrast, our proposed emitter-detector arrangement enables dual-slope measurements, enhancing depth sensitivity. OpenNIRScap [24] and NIRDuino [25] systems support higher channel counts, while our design is constrained by the two-emitter and two-detector configuration of the evaluation module. But it achieves a practical balance between accessibility, performance, and depth sensitivity.

The main limitation of the present system arises from the built-in detector package of the MAXM86146 module, which contains two fixed photodiodes separated by 2 mm. Since the detectors' configuration is not adjustable; this leads to two notable constraints. First, larger detector separations would enhance the depth sensitivity of the dual-slope analysis, but this cannot be achieved with the current design. Second, such a fixed detector arrangement limits the source-detector configuration, resulting in a probe length of at least 6 cm. On the head, this large probing area increases the possibility of sampling nonuniform tissue structures due to skull curvature or variation in extracerebral layer thicknesses. Since dual-slope analysis assumes equal extracerebral contributions across channels, such heterogeneity can introduce error and reduce the precision of absolute StO₂ estimation. Despite these limitations, the results are promising and demonstrate the potential of capturing changes in cerebral hemodynamics and oxygenation with this compact system. Future studies will include using the device for neonatal brain monitoring with a larger sample size.

5. Conclusion

This study presents the design, implementation, and validation of a low-cost, wearable NIRS system built from a battery-operated commercial optical biosensing module. The core contribution of this study is to validate the feasibility of repurposing a commercial optical biosensing evaluation module (MAXM86146EVSYS) for reliable NIRS-based cerebral oxygen monitoring. A major advantage of this approach is that the required modifications are minimal, making the system easy to assemble and adapt by non-experts. The device has reliable performance across multiple metrics, including signal quality, temporal stability, enhanced depth sensitivity, and physiological responsiveness. Layered blood phantom measurements confirmed its ability to detect signals beneath varying superficial layer thicknesses, while in vivo experiments validated its sensitivity to cerebral hemodynamic changes associated with vascular and respiratory challenges. These findings support the system's potential as a frugal neuromonitoring solution for neonates, particularly in low-resource environments.

Acknowledgments

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Data availability

The data cannot be made publicly available upon publication because they are not available in a format that is sufficiently accessible or reusable by other researchers. The data that support the findings of this study are available upon reasonable request from the authors. The data-processing code and 3D-printable probe design files used in this study are provided with this submission.

Supplementary data 1 available at <https://doi.org/10.1088/2515-7647/ae4cdf/data1>.

Data processing MATLAB codes available at <https://doi.org/10.1088/2515-7647/ae4cdf/data2>.

Probe 3D model: Base available at <https://doi.org/10.1088/2515-7647/ae4cdf/data3>.

Probe 3D model: PCB holder 1 available at <https://doi.org/10.1088/2515-7647/ae4cdf/data4>.

Probe 3D model: Lid available at <https://doi.org/10.1088/2515-7647/ae4cdf/data5>.

Probe 3D model: PCB holder 2 available at <https://doi.org/10.1088/2515-7647/ae4cdf/data6>.

Probe 3D model: LED holder available at <https://doi.org/10.1088/2515-7647/ae4cdf/data7>.

Probe 3D model: Strap holder available at <https://doi.org/10.1088/2515-7647/ae4cdf/data8>.

Ethical statement

All *in vivo* human measurements were performed in accordance with the Declaration of Helsinki and were conducted under study protocols approved by the Health Sciences Research Ethics Board of Western University (Protocol ID: 120391 and 107984), in accordance with the Tri-Council Policy Statement for research involving humans. Written informed consent was obtained from all subjects who participated in the study. The authors have confirmed that any identifiable participants in this study have given their consent for publication.

Conflict of interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization, S.S., D.W.H., and M.D.; Instrument development, S.S., and L.D.S.; Data Acquisition, S.S.; Data processing tools development, S.S., with input from R.E.; Data analysis, S.S., and R.E.; Statistical analysis, R.E.; Device characterization measurements, S.S.; Phantom measurement, S.S., and N.C.L.; Ethics approval, L.N.S., and M.D.; *In vivo* measurement, S.S., and L.N.S.; Supplemental *in vivo* data acquisition, L.N.S.; Resources, M.D.; Writing - original draft, S.S., and R.E., with input from N.C.L. and F.K.; Writing - review and editing, all authors; Visualization, S.S.; Supervision, M.D.; Project administration, M.D.; Funding acquisition, M.D.

ORCID iDs

Saeed Samaei  0000-0001-9399-1840

Rasa Eskandari  0009-0001-6702-9110

Natalie C Li  0000-0002-8263-3420

Farah Kamar  0000-0001-5913-0721

Lee D Sikstrom  0009-0002-2986-9546

Mamadou Diop  0000-0003-1244-7830

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